

A Resolution-Depth Grading of the Null Sector

Distinguishing Physical Blind Spots from Precision and Representation

Limits

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Abstract

The ignorance law states that exactly blind tangent directions receive zero Fisher length. This note refines that statement: the set of directions that *read* as null is not homogeneous. It stratifies by *resolution depth*—the order at which a direction’s information becomes positive as one refines along three achievable axes: sensors, arithmetic precision, and representation (model/symbol class). Three strata result, and they are operationally distinguishable. **Structural** null (depth ∞): exactly zero information at every precision and every symbol—a genuine physical blind spot, provably unrecoverable. **Precision** null (finite depth): information present but below the working arithmetic; a deeper number symbol (exact/arbitrary-precision arithmetic, equivalently an infinitesimal grading) separates it from true zero. **Representation** null (finite depth): information present in the data but inexpressible in the current symbol class; a richer class recovers it. All three are verified: a Fisher spectrum with eigenvalues 1, 10^{-20} , 0 is correctly split by 60-digit arithmetic into precision-limited and structural; a purely quadratic dependence invisible to a linear model ($R^2 = 0$) is recovered by a quadratic symbol ($R^2 = 0.9999$); and a structural null is shown invariant under every nonlinear feature map of the fixed data (residual 10^{-14}). This vindicates, in bounded form, the hypothesis that some of what we call null is a limit of our mathematical representation rather than a physical blind spot—while proving the complementary bound: symbol depth expands resolution *up to* the physical wall and never past it. The contribution is the grading and the classification, built from standard tools (valuations, arbitrary-precision arithmetic, feature refinement); it introduces no new primitive, and none is possible for crossing the wall.

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1 The question

The reconstruction and ignorance results settle what is recoverable from fixed observable data: the geometry up to gauge, and nothing inside an exact fibre. A natural and sharper question remains. When a direction reads as null, is that because *nature* withholds it (a sensor blind spot) or because *our representation is too coarse* to express information that is in fact present? If the two can be separated, then “improving mathematical resolution” has a precise meaning and a precise limit. We show they can be separated, and we locate the limit.

2 Resolution and depth

A null direction is null relative to a *resolution*: a triple $\rho = (\mathcal{S}, \pi, \mathcal{M})$ of an achievable sensor suite $\mathcal{S}$ (giving a Jacobian J), an arithmetic precision π , and a representation/model class $\mathcal{M}$. The Fisher form at resolution ρ is g_ρ^F , and its radical $\text{rad } g_\rho^F$ is the set of locally blind directions at that resolution.

Definition 1 (Resolution depth). For a direction $v \in \text{rad } g_{\rho_0}^F$ at a reference resolution ρ_0 , its resolution depth $\text{depth}(v)$ is the least refinement order along the three axes at which its information becomes positive:

$$\text{depth}(v) = \inf \{ k : g_\rho^F(v, v) > 0 \text{ for some resolution } \rho \text{ of order } k \},$$

with $\text{depth}(v) = \infty$ iff $g_\rho^F(v, v) = 0$ for every achievable ρ .

The three axes give three finite-depth mechanisms and one infinite-depth wall.

3 The three strata

Theorem 1 (Precision stratification). *Let v satisfy $g_{\rho_0}^F(v, v) = 0$ in the working (finite-precision) arithmetic. Under exact or arbitrary-precision arithmetic, either $g^F(v, v) = 0$ identically—structural, $\text{depth } \infty$ —or $0 < g^F(v, v) < \varepsilon_{\rho_0}$ —precision-limited, $\text{finite depth } \approx -\log_{10} g^F(v, v)$. The two are distinguishable, and only the latter is recovered by refining precision or sample size.*

Proof. Finite-precision evaluation collapses every value below the unit roundoff to zero, merging exact zeros with sub-threshold positives. Exact/arbitrary-precision evaluation does not: it returns the true value, which is either 0 or positive. Positivity is preserved under refinement and, by the finite-sample theorem, a positive Fisher eigenvalue is estimable at rate $n^{-1/2}$; an exact zero is not made positive by any amount of data. $\square$

Remark 1 (Verification). For eigenvalues $1, 10^{-20}, 0$: float64 reports $1, 0, 0$ (the machine epsilon 2.2×10^{-16} buries both small values), while 60-digit arithmetic reports $1, 10^{-20}, 0$, assigning depths $0, 20, \infty$. The “deeper number symbol” is exactly arbitrary-precision arithmetic; its idealization is the infinitesimal (a nonzero length below every standard positive real), i.e. a valuation grading of the radical.

Theorem 2 (Representation stratification). *Let $\mathcal{M} \subset \mathcal{M}'$ be nested representation classes and v a direction with $g_{\mathcal{M}}^F(v, v) = 0$ but $g_{\mathcal{M}'}^F(v, v) > 0$. Then v is representation-limited, not structural: the information is present in the data and is recovered by the richer symbol class.*

Proof. $g_{\mathcal{M}}^F$ is the pullback of the distinguishability form through the features of $\mathcal{M}$; a dependence orthogonal to $\mathcal{M}$ ’s feature span contributes zero there and positive in $\mathcal{M}'$ once the missing feature is included. The data are unchanged; only the expressible directions grow. $\square$

Remark 2 (Verification). With $y = x^2 + \text{noise}$ and symmetric x , the linear class $\{x, 1\}$ gives $R^2 = 0.0000$ —the dependence is invisible and reads as null—while the quadratic class $\{x^2, x, 1\}$ gives $R^2 = 0.9999$. This is the precise, benign form of “symbol-depth lack”: the null was an artifact of a coarse representation, and one additional symbol (x^2) dissolves it.

Theorem 3 (The wall: structural null is symbol-invariant). *If $\text{depth}(v) = \infty$ then for the fixed observable data F , every measurable function A of the data satisfies $A(F(x)) = A(F(x'))$ whenever $x' - x \in \text{span}(v)$, at every precision and in every representation class. No refinement of precision or symbol recovers v .*

Proof. Depth ∞ means v lies in $\ker J$ for every achievable sensor suite, so F is constant along v ; by the universal factorization (kernel-pair) theorem every observable map is fibre-constant, hence $A \circ F$ is constant along v . Precision and representation act on functions of F , not on F itself, so they cannot separate what F already identifies. $\square$

Remark 3 (Verification). For a latent difference of norm 1.32 lying in the unreachable directions of the sensor suite, every nonlinear feature map of the data— $z, z^2, z^3, \sin z, e^z$ —gives identical outputs to 10^{-14} . The physical wall is crossed by no symbol.

4 The classification, and what “more resolution” can do

Corollary 1 (Trichotomy of the null). *Every null direction is exactly one of:*

<i>stratum</i>	<i>depth</i>	<i>remedy</i>
<i>structural (physical blind spot)</i>	∞	<i>none—provably unrecoverable</i>
<i>precision-limited</i>	<i>finite (arithmetic/data)</i>	<i>exact/high precision, more samples</i>
<i>representation-limited</i>	<i>finite (symbol class)</i>	<i>a richer model/feature class</i>

decided by two tests: recompute the Fisher value in exact arithmetic (separates structural from precision), and enlarge the representation class (separates structural from representation).

This answers the motivating question. “Improving mathematical resolution” is real and has two concrete forms—a deeper number symbol (arbitrary precision / infinitesimal grading) and a deeper representation (feature/model refinement)—and each recovers a genuine, previously-null stratum. It also has an exact limit: neither can move a depth- ∞ direction, because that is not a property of our symbols but a theorem about the data. Symbol depth expands the resolved sector up to the physical wall and stops there.

Remark 4 (Relation to the ignorance law). This refines, and does not contradict, the ignorance law. That law assigns zero Fisher length to blind directions *at a fixed resolution*. Here the radical itself is graded: among directions of zero length at resolution ρ_0 , the structural ones have length zero at *every* resolution, while the precision- and representation-limited ones acquire positive length under refinement. “Zero Fisher length” is resolution-relative; depth makes the distinction precise.

5 Non-claims

Non-claim 1 (No crossing the wall). No number system, symbol, or representation recovers a structural (depth- ∞) null. The wall is the reconstruction/ignorance theorem, not a limitation of current notation.

Non-claim 2 (Depth is resolution-relative). Finite depths are defined relative to the *achievable* precision, sample, and representation classes. A direction unreachable within the achievable classes is structural for that program even if a physically inaccessible super-class would reach it.

Non-claim 3 (No new primitive). The grading uses standard tools—valuations/infinitesimals for precision depth, feature/model refinement for representation depth, the kernel-pair theorem for the wall. The contribution is the stratification and its tests, not a new mathematical object; and, per Theorem 3, no new object can change the wall.

6 Conclusion

The null sector is not one thing. It is graded by resolution depth into a physical stratum that no mathematics can open and two finite-depth strata that “deeper” mathematics genuinely can: one opened by a deeper number symbol (exact precision, the infinitesimal), one by a deeper representation (a richer symbol class). The hypothesis that some null is caused by the coarseness of our abstraction is therefore correct—precisely, and only, for the finite-depth strata—and the same grading proves the physical stratum is beyond every symbol. That double statement, a real gain in resolution together with its exact boundary, is the honest and publishable form of the idea.

References

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